PART A

StringBuffer, StringBuilder & Wrapper Classes

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1. Text Editor Simulator

Create a program that simulates a simple text editor using **StringBuilder**.
Support operations:

- append text
- insert at position
- delete range
- display final text

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2. String Compressor

Given a string like:

```
aaabbccdd → a3b2c2d2
```

Use **StringBuilder** to build the compressed string.

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3. Reverse Words in Sentence

Input:

```
Java is powerful
```

Output:

```
avaJ si lufrewop
```

Use **StringBuffer** to reverse each word.

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4. Large Data Concatenation

Simulate merging **10,000 strings** using:

- `String`
- `StringBuilder`
  Compare performance.

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5. Input Type Validator

Accept values as `String` and use **Wrapper classes** to:

- detect if input is Integer, Double, or Boolean
- convert and store accordingly.

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10. Banking Transaction Analyzer

Transactions are stored as strings:

```
"5000", "12000", "3000", "15000"
```

Tasks:

- Convert to Integer
- Find highest transaction
- Calculate average

PART B

Date Time API, Regular Expressions & Formatting

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1. Age Calculator

Accept date of birth as string and calculate **current age** using **Date Time API**.

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2. Event Countdown

Given a future date, calculate how many **days are left** for the event.

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3. Attendance Formatter

Accept attendance date in format:

```
2026-01-08
```

Display it as:

```
08-Jan-2026
```

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4. Email Validator

Create a program that validates email IDs using **Regular Expressions**.

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5. Password Strength Checker

Validate password rules using regex:

- minimum 8 characters
- at least 1 uppercase
- at least 1 digit
- at least 1 special character

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6. Mobile Number Validator

Check if the mobile number is valid:

- must start with 6–9
- must be 10 digits

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7. Currency Formatter

Accept salary as a number and display it in:

- Indian format
- US format

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8. Invoice Date Generator

Generate today's date and display in formats:

- `dd-MM-yyyy`
- `MMMM dd, yyyy`
- `EEEE` (day name)

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9. Extract Numbers from Text

Given text:

```
Order 45 costs 1200 rupees
```

Use regex to extract all numbers.